Pranshu Tripathi

Software Engineer

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EXPERIENCE

Schrödinger

Senior Developer I *January 2026 - Present*

- Project lead of Maestro Assistant.
- Developed a Semantic Panel Search system to seamlessly find relevant Maestro Panels for scientific tasks, successfully validating the architecture for production deployment
- Collaborating on architecture design of Notification and Authentication services for Maestro.

Developer II *July 2024 - December 2025*

- Spearheaded R&D for 'Maestro Assistant', an LLM-driven research and agentic tool capable of performing actions in Maestro as per user request. Engineered a RAG (Retrieval-Augmented Generation) pipeline using Vector Embeddings to provide a seamless way to search through Schrödinger's extensive documentation.
- Implemented non-bonded interaction calculation library; a modular C++ library with caching. It reduced the calculation time by 300-400x; cut 10-min tasks to 30s.

Developer I *June 2023 - June 2024*

- Refactored 2000+ lines of legacy code using Factory and Strategy patterns, improving extensibility for handling diverse Job Server outputs.
- Enhanced export functionality for new movie presets (e.g., View Ligand Docking) from Maestro to PyMOL, decreasing manual visualization setup time by 60% and enhancing automated workflow efficiency.

Intern - Software Development *January 2023 - May 2023*

- Created an adapter for Schrödinger's structure class that removed the m2io dependency and enabled a property-writing mechanism with 3x greater flexibility, gaining recognition from Scientific Infrastructure team.
- Implemented a robust Find Toolbar widget following MVC pattern and command pattern with python and C++ inter communication.
- Delivered a robust preset management feature in 2 months—30% faster than average—using Factory and Adapter design patterns.

PROJECTS

EventHive

Deployed a ticketing platform on GKE handling 3,000+ RPS, publishing NPM packages to streamline microservice integration across 5 services

PlotCortex

Built a semantic search engine for the IMDB dataset using Sentence Transformers Embeddings and Vector DB Qdrant.

CarbonHero

Built a leaderboard-based Flutter app for GDSC Solution Challenge 2021 to track user carbon footprint, integrating Firebase for real-time updates and auth

Gun Shot Sound Classification (Minor Project IIIT-A)

Built a gunshot sound classification model using MFCC features and machine learning algorithms like LSTM and CNN to achieve high accuracy in identifying gunshot sounds from audio data.

Cardiovascular Disease Detection Analysis on Phonocardiogram Signals (Major Project IIIT-A)

Built CNN & LSTM models to analyze their accuracies in detecting CVD from PCG data, with pre processing of signals using DWT as a major project work at IIIT-A.

EDUCATION

IIIT Allahabad (B.Tech)

Electronics and Communication
Aug 2019 - May 2023

- **GPA: 9.16/10**
- **Batch Rank: 4**
- **Awards**
  - Merit-cum-Means Scholarship for Academic Excellence 2020, 2021, 2022
  - Cleared J.E.E Mains 2019
  - Cleared J.E.E Advanced 2019

COURSEWORK

Computer Science
Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering

Mathematics
Linear Algebra, Probability and Statistics, Calculus

Electronics and Communication
Microprocessors and Microcontrollers, Digital System Design, Digital Signal Processing and Communication

SKILLS

Programming Languages

- C++, Python, Java, JavaScript, TypeScript, SQL, Shell Scripting, Terraform

Framework & Libraries

- Spring, Express.js, React, Next.js, Qt, Hibernate, PostgreSQL, SIP (Python-C++), PyQt6, BULLJS

Cloud & DevOps

- Docker, Kubernetes (K8s), GitHub Workflows, Travis CI, AWS (Elastic Beanstalk, RDS, ElastiCache), GCP, GKE, G Cloud SQL, NATs Streaming

Generative AI Technologies

- LiteLLM, Vertex AI, Hugging Face, Qdrant Vector DB, LangChain, DSPY, MCP, RAG

Machine Learning & Specialized Skills

- CNNs, LSTMs, Transformers, Graph DB, IoT Systems, Classification Models, Speech to Text recognition

VOLUNTEERING

- Coordinator - IoT Wing, IIIT Allahabad
- Senior Member - Literature Society, IIIT Allahabad

LANGUAGES

English, Hindi, Marathi, German